FLORIDA RESIDENTIAL SWIMMING POOL DESIGN

LEGEND

GN1 - COVER & GENERAL NOTES
S1 - POOL LAYOUT
D1 - POOL DETAILS 1
D2 - POOL DETAILS 2

NOTE: SHEETS TO SCALE ON 24" x 36" PAPER

811 Know what's below. Call before you dig.

CALL BEFORE YOU DIG

CALL LOCAL 811 TO OBTAIN LOCATION OF UNDERGROUND UTILITIES A MIN. OF 3 DAYS IN ADVANCE BEFORE YOU DIG.

Table 13.1 - Reinforcing Bar Development and Splice Lengths for f'c = 3000 psi and fy= 60 ksi

Bar Size	Tension Development		Class "B" Splice		Std. 90 deg. Hook		Compression Bars	
	"Top Bar Ldt (in.)"	"Other Bar Ldo (in.)"	"Top Bar Lbt (in.)"	"Other Bar Lbo (in.)"	"Embed. Ldh (in.)"	"Leg Length Lh (in.)"	"Develop. Ldc (in.)"	"Splice (in.)"
#3	12	12	16	16	6	6	8.00	12
#4	15	12	19	16	7	8	9	15
#5	18	14	24	18	8	10	12	19
#6	22	17	29	22	10	12	14	23
#7	32	25	42	32	12	14	17	26
#8	37	28	48	37	13	16	19	30
#9	46	35	60	46	15	19	21	34
#10	57	44	74	57	17	22	24	38
#11	68	52	88	68	19	24	27	42
#14	93	72	--	---	32	31	32	---
#18	151	116	-	---	43	41	43	---

NOTES:
1. STRAIGHT DEVELOPMENT AND CLASS "B" SPLICE LENGTHS SHOWN IN ABOVE TABLE ARE BASED ON UNCOATED BARS ASSUMING CENTER-TO-CENTER BAR SPACING OF I2" WITHOUT TIES OR STIRRUPS AND BAR CLEAR COVER OF 2". NORMAL WEIGHT CONCRETE WITH NO TRANSVERSE REINFORCING ARE BOTH ASSUMED. NO EXCESS REINFORCING IS ASSUMED.
2. STANDARD 90 DEGREE HOOK EMBEDMENT LENGTHS ARE BASED ON BAR SIDE COVER OF 2.5" AND BAR END COVER OF 2" WITHOUT TIES AROUND HOOK.
3. TABLE DOES NOT REFLECT SPECIAL SEISMIC CONSIDERATIONS FROM ACI 318-05, CHAPTER 21.

CONSTRUCTION CODE REQUIREMENTS

THESE PLANS WERE PREPARED AND SHALL COMPLY WITH THE

A. FLORIDA STATE:
2023 FLORIDA BUILDING CODE 8TH EDITION, RESIDENTIAL, CHAPTER NO. 45, SECTION: R4501
2023 FLORIDA BUILDING CODE 8TH EDITION, ENERGY CONSERVATION, CHAPTER NO. 4
SECTION: R403.10 POOLS AND PERMANENT SPA ENERGY CONSUMPTION (MANDATORY)
SECTION: R403.11 PORTABLE SPA (MANDATORY)
SECTION: R403.12 RESIDENTIAL POOL AND PERMANENT SPA
2023 FLORIDA STATUTES, CHAPTER NO. 515 RESIDENTIAL SWIMMING POOL SAFETY ACT

B. FEDERAL GOVERNMENT:
2014 ANSI/ APSP/ ICC-3: AMERICAN NATIONAL STANDARD FOR PERMANENTLY INSTALLED RESIDENTIAL SPAS SWIM SPAS R4501.6.1
2012 ANSI/ APSP/ ICC-5: AMERICAN NATIONAL STANDARD FOR RESIDENTIAL INGROUND SWIMMING POOLS R4501.6.1
2020 ANSI/APSP/ICC-7: AMERICAN NATIONAL STANDARD FOR SUCTION ENTRAPMENT AVOIDANCE IN SWIMMING POOLS, WADING POOLS, SPAS, HOT TUBS AND CATCH BASINS R4501.6.1, R4501.6.3, R4501.6.6
2013 ANSI/ASHRAE-15: SAFETY STANDARDS FOR REFRIGERATION SYSTEMS.
2012 ANSI/ APSP/ ICC 4-12: AMERICAN NATIONAL STANDARD FOR ABOVE-GROUND/ ON-GROUND RESIDENTIAL SWIMMING POOLS R4501.6.1.
2013 ANSI/ APSP/ ICC 6-13: AMERICAN NATIONAL STANDARD FOR RESIDENTIAL PORTABLE SPAS AND SWIM SPAS
2020 NATIONAL ELECTRICAL CODE (NEC) ARTICLE NO. 680

IMPORTANT NOTE: POOL CONTRACTOR IS RESPONSIBLE FOR CONFORMING TO ALL ABOVE LISTED CODE REQUIREMENTS AS WELL AS ANY ADDITIONAL REQUIREMENTS PER LOCAL MUNICIPALITY THAT MAY BE MORE STRINGENT THAN THE ABOVE LISTED CODES REQUIREMENTS.

GENERAL POOL / SPA REQUIREMENTS:

THE POOL CONTRACTOR IS RESPONSIBLE FOR ENSURING ALL COMPONENTS OF THE POOL ARE IN ACCORDANCE WITH THE FLORIDA BUILDING CODE, AND ALL CONSTRUCTION SHALL MEET ALL APPLICABLE CODES INCLUDING PLUMBING, ELECTRICAL AND GAS.

RESIDENTIAL SWIMMING POOL IS NOT DESIGNED FOR DIVING. LADDER OR STAIRS ARE TO BE PROVIDED. ENTRY/ EXIT REQUIRED AT THE SHALLOW END AND DEEP END IF OVER 5 FEET DEEP. ACCEPTABLE OPTIONS ARE STAIRS (10" MINIMUM TREAD WITH 240 SQUARE INCH MINIMUM AREA, 12" RISER WITH INTERMEDIATE TREADS AND RISERS UNIFORM), LADDERS, UNDERWATER SEATS, AND SWIM-OUTS (MAXIMUM 20" BELOW WATERLINE).
OUTDOOR SWIMMING POOLS ARE TO HAVE BARRIERS THAT COMPLY WITH THE FLORIDA BUILDING CODE.
ALL WALL SURROUNDING INDOOR SWIMMING POOLS SHALL COMPLY WITH THE FLORIDA BUILDING CODE.
FINAL ELECTRICAL, AND BARRIER CODE, INSPECTIONS SHALL BE COMPLETED PER LOCAL MUNICIPALITY GUIDELINES

SPECIAL SPA REQUIREMENTS:

MAXIMUM WATER DEPTH 4', MAXIMUM SEAT DEPTH 28".
FLOOR SLOPE 1:12
MIN. TREAD 10" X 12" MAX. RISER.
INTERMITTENTLY THE SPA SHALL HAVE A ONE HOUR TURNOVER, IF CONTINUOUS A SIX HOUR TURN OVER.
MAXIMUM TEMPERATURE 104 DEGREES.

GENERAL NOTES:

1. THIS DESIGN IS LIMITED EXCLUSIVELY TO THE POOL SHELL. ANY ADJACENT DECKING OR RELATED STRUCTURES AND EQUIPMENT ARE OUTSIDE OF THE SCOPE OF THIS DESIGN AND ARE TO BE DESIGNED BY OTHERS. THE ENGINEER OF RECORD AND VITO GIRONDA ENGINEERING, LLC ARE NOT RESPONSIBLE FOR INFORMATION PROVIDED BY OTHERS.

2. DUE TO LOCAL WATER TABLE ELEVATION, DAMAGE MAY OCCUR TO THE POOL STRUCTURE IF THE POOL IS DRAINED. THE OWNER SHALL CONTACT THE POOL BUILDER AND ENGINEER IN WRITING FOR SPECIFIC INSTRUCTIONS PRIOR TO DRAINING THE POOL. HYDROSTATIC UPLIFT PRESSURE MUST BE DETERMINED BEFORE DRAINING MAY OCCUR. THE ENGINEER ASSUMES NO LIABILITY FOR ACTIVITIES INVOLVING DRAINING THE POOL.

3. THE PROPERTY BOUNDARY SHOWN HEREON IS BASED ON PUBLIC RECORDS AND NOT THE RESULT OF A BOUNDARY SEARCH BY A LICENSED SURVEYOR. NO REPRESENTATION IS MADE TO ITS ACCURACY AND THE ENGINEER SHALL NOT BE LIABLE FOR DISCREPANCIES, DISPUTES, OR LEGAL MATTERS ARISING FROM THE USE OF THESE PLANS FOR BOUNDARY-RELATED PURPOSES. IT IS REQUIRED THAT A LICENSED SURVEYOR BE CONSULTED TO CONDUCT A COMPREHENSIVE BOUNDARY SURVEY TO ASCERTAIN ACCURATE AND LEGALLY DEFENSIBLE PROPERTY LINES. THE ENGINEER ASSUMES NO RESPONSIBILITY FOR THE POOL CONSTRUCTION IN EASEMENTS OR ON REQUIRED SETBACK AREAS. POOL CONTRACTOR AND/OR OWNER SHALL VERIFY LAYOUT AND DIMENSIONS PRIOR TO CONSTRUCTION.

4. THE ENGINEER WILL NOT BE RESPONSIBLE FOR THE CONTRACTOR'S MEANS, METHODS, TECHNIQUES, SEQUENCES OF PROCEDURES OR CONSTRUCTION, OR THE SAFETY PRECAUTIONS AND PROGRAMS INCIDENT THERETO, AND THE ENGINEER WILL NOT BE RESPONSIBLE FOR THE CONTRACTOR'S FAILURE TO PERFORM THE WORK IN ACCORDANCE WITH THE CONTRACT DOCUMENTS AND OR LOCAL CODES OR REGULATORY STANDARDS OF PRACTICE.

5. THE CONTRACTOR MUST PROTECT EXISTING STRUCTURES AND UTILITIES DURING CONSTRUCTION. THE ENGINEER ACCEPTS NO RESPONSIBILITY FOR THE SAFETY OF EXISTING STRUCTURES, UTILITIES, OR SEPTIC SYSTEMS AND IS LIMITED TO THE STRUCTURAL DESIGN OF THE POOL SHELL. THE CONTRACTOR SHALL CONTACT 811 MINIMUM OF 3 DAYS PRIOR TO DIGGING ACTIVITIES.

6. THE ENGINEER ASSUMES NO LIABILITY FOR UTILITY WORK, SURROUNDING SITE WORK, OR PROPER EROSION AND SEDIMENT CONTROL MEASURES. THE CONTRACTOR SHALL ENSURE ALL WORK IS IN COMPLIANCE WITH LOCAL REGULATIONS.

7. THE POOL AREA SHALL BE FENCED DURING AND AFTER CONSTRUCTION. NO OPENING SHALL BE PERMITTED WHILE THE CONSTRUCTION SITE IS LEFT UNATTENDED PER FBC REQUIREMENTS.

8. POOL DESIGN IS ACCEPTABLE FOR MAXIMUM 25'-0" X 25'-0" RESIDENTIAL POOL WITH MAXIMUM DEPTH OF 5'-0". SHAPE VARIATIONS WITHIN MAXIMUM DEFINED ENVELOPE PERMITTED WHEN COORDINATED WITH THE OWNER AND ENGINEER, IN WRITING.

9. POOL STRUCTURE WALL SHALL BE NOT BE BUILT WITHIN THE ANGLE OF REPOSE OF ANY ADJACENT STRUCTURE WITHOUT SPECIAL AOR ENGINEERING.

10. THIS DESIGN IS NOT VALID FOR CONSTRUCTION SEAWARD OF THE COASTAL CONSTRUCTION CONTROL LINE.

11. IN THE EVENT OF CONFLICT WITH THESE PLANS AND THE FLORIDA BUILDING CODE, THE BUILDING CODE SHALL TAKE PRECEDENCE, UNLESS OTHERWISE APPROVED BY THE ENGINEER OF RECORD IN WRITING.

12. ALL COMPONENTS SHALL COMPLY WITH THE NSPI SPECIFICATIONS AND SHALL HAVE NSPI APPROVAL EFFECTIVE THE DATE OF THE PERMIT APPLICATION. INSTALLATION AND COMPONENT SELECTION SHALL COMPLY WITH THE MOST RECENT EDITION OF THE FLORIDA BUILDING CODE AND NSPI-8 LATEST EDITION.

13. POOL ENCLOSURES, CHILD BARRIERS, FENCES, ALARMS, AND COMPONENTS SHALL BE DESIGNED BY OTHERS. ALL COMPONENTS SHALL COMPLY WITH NSPI SPECIFICATIONS AND HAVE NSPI APPROVAL EFFECTIVE THE DATE OF THE PERMIT. THE POOL SHALL BARRIER BE COMPLIANT WITH THE REQUIREMENTS OF FLORIDA BUILDING CODE 2023 EDITION RESIDENTIAL, CHAPTER 45, SECTION R4501.17

14. GLAZING IN WALLS, ENCLOSURES OR FENCES CONTAINING, FACING OR ADJACENT TO HOT TUBS, SPAS, WHIRLPOOLS, SAUNAS, STEAM ROOMS, BATHTUBS, SHOWERS AND INDOOR OR OUTDOOR SWIMMING POOLS WHERE THE BOTTOM EXPOSED EDGE OF THE GLAZING IS LESS THAN 60 INCHES (1524 MM) MEASURED VERTICALLY ABOVE ANY STANDING OR WALKING SURFACE SHALL BE CONSIDERED TO BE A HAZARDOUS LOCATION. THIS SHALL APPLY TO SINGLE GLAZING AND EACH PANE IN MULTIPLE GLAZING.

15. ALL DOORS AND WINDOWS PROVIDING DIRECT ACCESS FROM THE DWELLING TO THE POOL SHALL BE EQUIPPED WITH AN EXIT ALARM COMPLYING WITH FBC R4501.17.1.9.1 AND UL-2023 THAT HAS MINIMUM SOUND PRESSURE RATING 85 DB AT 10 FEET, DOOR AND WINDOW ALARM DEACTIVATION SWITCH SHALL BE LOCATED AT LEAST 54" ABOVE THRESHOLD FOR ACCESS. ALL DOORS PROVIDING DIRECT ACCESS FROM THE DWELLING TO THE POOL SHALL BE EQUIPPED WITH SELF-CLOSING, SELF-LATCHING DEVICES AT 54" ABOVE THE THRESHOLD THAT MEETS REQUIREMENTS OF FBC R4501.17.1.9 (2).

16. HANDHOLD SHALL BE PROVIDED AROUND THE POOL EDGE IN ANY AREA WHERE THE WATER DEPTH EXCEEDS 4 FT. AS PER ANSI/ APSP -5 SECTION 17.1

17. ALL POOLS WHETHER PUBLIC OR PRIVATE SHALL BE PROVIDED WITH A LADDER OR STEPS IN THE SHALLOW END WHERE WATER DEPTH EXCEEDS 24 INCHES. IN PRIVATE POOLS WHERE WATER DEPTH EXCEEDS 5 FEET, THERE SHALL BE LADDERS, STAIRS OR UNDERWATER BENCHES/SWIMOUTS IN THE DEEP END. WHERE MANUFACTURED DIVING EQUIPMENT IS TO BE USED, BENCHES OR SWIMOUTS SHALL BE RECESSED OR LOCATED IN A CORNER. ALL MEANS OF EGRESS SHALL BE IN ACCORDANCE WITH THE 8TH/ EDITION FLORIDA BUILDING CODE.

18. DESIGN FOR THIS NEW CONSTRUCTION IS VALID FOR ONE YEAR FROM THE DATE OF ISSUE OR UPON ADOPTION OF NEW LAWS PERTAINING TO RESIDENTIAL POOL CONSTRUCTION HAS BEEN REVISED BY THE AUTHORITY HAVING JURISDICTION.

19. THE CONTRACTOR AGREES THAT THEY SHALL ASSUME SOLE AND COMPLETE RESPONSIBILITY FOR JOB SITE CONDITIONS DURING THE COURSE OF CONSTRUCTION OF THIS PROJECT, INCLUDING SAFETY OF ALL PERSONS AND PROPERTY; THAT THIS REQUIREMENT SHALL APPLY CONTINUOUSLY AND NOT BE LIMITED TO NORMAL WORKING HOURS, AND THAT THE CONTRACTOR SHALL DEFEND, INDEMNIFY, AND HOLD THE OWNER AND ENGINEER HARMLESS FROM ANY AND ALL LIABILITY, REAL OR ALLEGED, IN CONNECTION WITH THE PERFORMANCE OF WORK ON THIS PROJECT, EXCEPTING FOR LIABILITY ARISING FROM THE SOLE NEGLIGENCE OF THE OWNER, OR THE ENGINEER, OR ANY PUBLIC AGENCY.

SITE DEMOLITION

1. PROJECT SITE TO BE CLEAR OF ANY AND OR ALL OBSTRUCTION PRIOR TO START OF CONSTRUCTION (BY OWNER). CALL 811 A MINIMUM OF 3 DAYS PRIOR TO DIGGING ON SITE. CONTRACTOR RESPONSIBLE FOR LOCATING, MARKING, AND PROTECTING ALL EXISTING UTILITIES AND STRUCTURES PRIOR TO CONSTRUCTION ACTIVITIES.

2. OWNER TO PROVIDE ANY OR ALL UTILITIES TO COMPLETE CONSTRUCTION AND ENSURE PROPER OPERATION (DESIGNED BY OTHERS). ALL EXISTING PLUMBING / PIPING LINES, ELECTRICAL, TO BE ABANDONED OR REMOVED, IF NECESSARY (DESIGNED BE OTHERS).

3. CONTRACTOR SHALL BE RESPONSIBLE FOR DEMOLITION WORK WHICH SHALL BE EXECUTED IN CONFORMANCE WITH ALL CODES AND ORDINANCES AS SET FORTH BY ALL GOVERNING AUTHORITIES. CONTRACTOR IS RESPONSIBLE FOR PROVIDING AND IMPLEMENTING ALL NECESSARY EROSION AND SEDIMENT CONTROL MEASURES BASED ON BMPS AND ALL USEPA AND LOCAL REGULATIONS (DESIGNED BY OTHERS).

4. ALL INFORMATION REGARDING EXISTING CONDITIONS IS BASED ON VISUAL INSPECTION AND MAY NOT REFLECT ACTUAL FIELD CONDITIONS. UPON DISCOVERY OF ANY DISCREPANCIES BETWEEN DRAWINGS DEPICTING EXISTING CONDITIONS OR UPON DISCOVERY OF UNKNOWN CONDITIONS DETRIMENTAL TO THE COMPLETION OF THE WORK AS INDICTED ON THE DRAWINGS, THE CONTRACTOR SHALL IMMEDIATELY NOTIFY THE ENGINEER, IN WRITING, OF THE CONDITION IN QUESTION BEFORE PROCEEDING.

5. EXCAVATE, AS NECESSARY, FOR ALL UNDERGROUND PLUMBING WHICH INCLUDES ALL DRAIN AND RETURN LINES FOR POOL. POOL PLUMBING TRENCH TO ALLOW ALL PIPES TO BE A MIN. OF 2FT BELOW FINISHED GRADE.

6. ALL DEMOLITION AND CONSTRUCTION TO BE STAKED PRIOR TO CONSTRUCTION. NOTIFY OWNER TO REVIEW LAYOUT STAKING AND APPROVE PLACEMENT, PRIOR TO PROCEEDING WITH CONSTRUCTION ACTIVITIES OWNER TO PROVIDE WRITTEN APPROVAL OF STAKING LOCATION PRIOR TO COMMENCING CONSTRUCTION. THE ENGINEER ASSUMES NO LIABILITY FOR THE STAKING PROCESS (OR LACK THEREOF) AND THE FINAL LOCATION OF THE PROPOSED POOL.

7. FOLLOWING CONSTRUCTION, CONTRACTOR TO ENSURE ALL DISTURBED AREAS ARE FINISHED WITH SLOPE TO PROMOTE POSITIVE DRAINAGE.

STRUCTURAL NOTES:

1. ALL CONCRETE SHALL CONFORM TO THE BUILDING CODE REQUIREMENTS FOR REINFORCED CONCRETE (ACI 318), THE STANDARD SPECIFICATION FOR STRUCTURAL CONCRETE IN BUILDINGS (ACI 301) AND ALL LOCAL BUILDING CODES. ALL CONCRETE WORK SHALL BE AS SPECIFIED AND RECOMMENDED BY ACI FIELD REFERENCE MANUAL SP-15. SPECIAL COLD WEATHER (ACI 306) OR HOT WEATHER (ACI 305) CONCRETING PRACTICES SHALL BE UTILIZED WHENEVER APPROPRIATE. ALL SHOTCRETE SHALL ALSO CONFORM TO THE REQUIREMENTS OF ACI 506R-05 GUIDE TO SHOTCRETE.

2. SHOTCRETE AND GUNITE SHALL HAVE A MAXIMUM 3/8" STONE AGGREGATE AND CAST IN PLACE CONCRETE SHALL HAVE A MAXIMUM 1" STONE AGGREGATE. EACH SHALL HAVE A MINIMUM COMPRESSIVE STRENGTH OF 3,000 PSI AT 28 DAYS.

3. ALL REINFORCING BARS SHALL BE DEFORMED BARS CONFORMING TO REQUIREMENTS OF ASTM SPECIFICATION A615, GRADE 60 UNLESS OTHERWISE NOTED. ALL REINFORCING IN POOL SHELL SHALL BE TIED AT EACH CROSSING

4. REINFORCING SHALL BE SECURELY TIED IN ITS PROPER PLACE BEFORE AND DURING POURING OPERATIONS USING APPROVED CHAIRS AND SPACERS AS REQUIRED PER ALL APPLICABLE STANDARDS.

7. PROVIDE CLEARANCES FROM FACES OF CONCRETE TO REINFORCEMENT AS FOLLOWS UNLESS OTHERWISE SPECIFIED:

- SLABS ON FILL.......3"
- BEAMS.................1 1/2"
- FOOTINGS............3"
- WALLS.................2" EXTERIOR FACE, 2" INTERIOR FACE
- PIERS..................2" EXTERIOR FACE

8. DO NOT CUT OR DISPLACE REINFORCING STEEL TO ACCOMMODATE INSTALLATION OF EMBEDDED ITEMS UNLESS APPROVED BY THE ENGINEER IN WRITING. COORDINATE INSTALLATION OF SLEEVES, PIPES AND CONDUIT WITH THE PLACING OF REINFORCING STEEL TO ENSURE THAT IT DOES NOT NEED TO BE CUT OR DISPLACED.

9. ALL HORIZONTAL SURFACES INTENDED FOR FOOT TRAFFIC SHALL RECEIVE A NON-SLIP AND IMPERVIOUS FINISH.

10. CONCRETE SURFACES EXPOSED TO VIEW SHALL BE SOUND AND DENSE. PULL TIE RODS TO THE FULL LENGTH OF CONES. WHERE PATCHING OF HONEYCOMBS AND VOIDS IS REQUIRED, IT SHALL BE DONE AS SOON AS POSSIBLE

11. AFTER THE REMOVAL OF THE FORMS. REPAIR AND PLUG TIE HOLES AND HONEYCOMBS WITH NON-SHRINK CEMENT BASED PRODUCTS ACCORDING TO MANUFACTURER'S INSTRUCTIONS.

12. ALL LAPS SHALL BE "B" SPLICES UNLESS NOTED OTHERWISE ON THE DRAWINGS. SPECIAL PLACEMENT AND SPLICING OF REINFORCEMENT USED IN SHOTCRETE APPLICATIONS SHALL CONFORM TO THE REQUIREMENTS OF IBC-2006 SECTION 1913.4.

13. ALL CONCRETE SHALL BE CURED FOR A MINIMUM OF SEVEN DAYS. ANY UNFINISHED SHOTCRETE WORK SHALL NOT BE ALLOWED TO STAND FOR MORE THAN 30 MINUTES UNLESS EDGES ARE SLOPED TO A THIN EDGE, UNLESS DETAILED OTHERWISE. PRIOR TO PLACING ADDITIONAL MATERIAL ADJACENT TO PREVIOUSLY APPLIED WORK, ALL EDGES SHALL BE CLEANED AND WETTED.

14. WHEREVER SLEEVES ARE INSERTED IN CONCRETE SLABS, BEAMS OR WALLS, THEY SHALL CONSIST OF PVC PIPE.

15. ALL CONCRETE REINFORCING WITHIN THE POOL AREA TO BE BONDED AND GROUNDED ELECTRICALLY IN ACCORDANCE WITH NATIONAL, STATE, AND LOCAL CODES, BY THE ELECTRICAL CONTRACTOR. THE ENGINEER NOT RESPONSIBLE FOR PROPER ELECTRICAL WORK.

16. CALCIUM CHLORIDE AND OR ADMIXTURES CONTAINING CHLORIDE SHALL NOT BE USED.

17. POOL SHELL AND ALL FOOTINGS ARE DESIGNED ASSUMING COMPETENT, NON-EXPANSIVE, SOILS OR APPROVED ENGINEERED FILL WITH CONTINOUS SUPPORT AND ALLOWABLE BEARING CAPACITY QA = 3,000 PSF, EFFECTIVE UNIT WEIGHT $\Gamma = 110$ PCF, $\Phi \geq 30°$, AND SEASONAL HIGH GROUNDWATER ≥ 12 IN BELOW SLAB SUBGRADE AT TIME OF PLACEMENT. IT IS THE RESPONSIBILITY OF THE CONTRACTOR TO INVESTIGATE, VERIFY, AND DOCUMENT THAT IN-PLACE SOILS AND GROUNDWATER MEET OR EXCEED THE DESIGN BASIS AS PART OF THE DESIGN.

18. IF SETTLING OR WEATHERING OF THE POOL DECK OCCURS THAT WOULD CAUSE STANDING WATER, THE ORIGINAL SLOPES SHALL BE RESTORED, OR CORRECTIVE DRAINS INSTALLED IMMEDIATELY WITH ACCESS PROHIBITED UNTIL CORRECTED (DESIGNED BY OTHERS).

19. REFER TO TABLE 13.1 FOR REINFORCING BAR DEVELOPMENT AND SPLICE LENGTHS.

20. EXISTING STRUCTURES ARE DEPICTED GRAPHICALLY. ALL DIMENSIONS OR ELEVATIONS SHOWN SHOULD BE FIELD VERIFIED. GENERAL CONTRACTOR, FABRICATOR AND ERECTOR SHALL DETERMINE WHO IS RESPONSIBLE FOR DETERMINING WHO WILL PROVIDE FIELD MEASUREMENTS. ALL FIELD MEASUREMENTS WILL BE REFLECTED ON SHOP DRAWINGS.

21. STRUCTURAL SPECIFICATIONS FOR ACRYLIC INSERTS USED IN THIS PROJECT, IF APPLICABLE, ARE TO BE PROPERLY DESIGNED BY OTHERS. THE ENGINEER OF RECORD ASSUMES NO RESPONSIBILITY FOR THE ACRYLIC INSERT OR THE INSTALLATION PROCESS USED.

22. FOR ALL WALKING SURFACES LOCATED MORE THAN 30" ABOVE GRADE, GUARDS SHALL BE PROVIDED PER FBC R312.

MECHANICAL, ELECTRICAL, AND PLUMBING NOTES:

1. WATER AND SANITARY SEWER SERVICE IS PROVIDED BY THE OWNER. ANY MODIFICATIONS ARE NOT PART OF THIS DESIGN.

2. ALL PIPING, EQUIPMENT, AND MATERIALS USED IN THE POOL SYSTEM SHALL CONFORM TO THE 2018 INTERNATIONAL SWIMMING POOL AND SPA CODE, THE 2020 NATIONAL ELECTRIC CODE, THE 2022 (REAFFIRMED) AMERICAN NATIONAL STANDARD FOR PUBLIC SWIMMING POOLS (ANSI/APSP/ICC-2) THE 8TH EDITION FLORIDA BUILDING CODE.

3. POOL MUST INCLUDE ALL COMPONENTS OF THE RECIRCULATION SYSTEM PER THE 2018 INTERNATIONAL SWIMMING POOL AND SPA CODE AND THE 8TH/ EDITION FLORIDA BUILDING CODE.

4. POOL PIPING SHALL BE SUPPLIED SO THE WATER VELOCITY WILL NOT EXCEED 10 FPS FOR PRESSURE PIPING AND 6 FPS FOR SUCTION PIPING.

5. ALL EQUIPMENT AND ASSOCIATED PIPING SHALL BE INSTALLED IN STRICT ACCORDANCE WITH MANUFACTURER'S STANDARDS.

6. ENTRAPMENT PROTECTION FOR SUCTION OUTLETS SHALL BE INSTALLED IN ACCORDANCE WITH THE REQUIREMENTS OF ANSI/APSP/ICC-7 & FBC R4501.6.6. BLOCKABLE DUAL OUTLETS (MAIN DRAINS) SHALL BE SEPARATED BY A MINIMUM OF 3 FEET MEASURED FROM CENTER TO CENTER OF THE SUCTION OUTLET FITTING ASSEMBLY (SOFA) OR LOCATED ON TWO (2) DIFFERENT PLANES I.E., ONE (1) ON THE BOTTOM AND ONE (1) ON THE VERTICAL WALL, OR ONE (1) EACH ON TWO (2) SEPARATE VERTICAL WALLS. SUCTION OUTLETS SHALL NOT BE INSTALLED IN SEATING AREAS. AN UNBLOCKABLE SOFA REQUIRES THAT THE SUCTION OUTLET FITTING ASSEMBLY (SOFA) BE CERTIFIED AS UNBLOCKABLE, AND BE DESIGNED BY THE MANUFACTURER AS UNBLOCKABLE, AND THE MANUFACTURER'S INSTRUCTIONS MUST STATE THE SOFA IS AUTHORIZED FOR USE AS AN UNBLOCKABLE IN ACCORDANCE WITH ANSI/ APSP/ ICC - 16 2017.

7. POOL CIRCULATING PUMPS SHALL BE EQUIPPED ON THE INLET SIDE WITH AN APPROVED-TYPE HAIR AND LINT STRAINER WHEN USED WITH A PRESSURE FILTER.

8. PUMP IMPELLERS, SHAFTS, WEAR RINGS AND OTHER WORKING PARTS SHALL BE OF CORROSION-RESISTANT MATERIALS.

9. COMPONENTS SHALL HAVE SUFFICIENT CAPACITY TO PROVIDE A COMPLETE TURNOVER OF POOL WATER IN 12 HOURS OR LESS.

10. ALL JOINTS, CONNECTIONS, AND VALVES SHALL BE MADE OF MATERIALS THAT ARE APPROVED IN THE FLORIDA BUILDING CODE, PLUMBING. VALVES LOCATED UNDER CONCRETE SLABS SHALL BE SET IN A PIT HAVING A LEAST DIMENSION OF FIVE PIPE DIAMETERS WITH A MINIMUM OF AT LEAST 10 INCHES AND FITTED WITH A SUITABLE COVER. ALL VALVES SHALL BE LOCATED WHERE THEY WILL BE READILY ACCESSIBLE FOR MAINTENANCE AND REMOVAL.

11. APPROVED SURFACE SKIMMERS ARE REQUIRED AND SHALL BE INSTALLED IN STRICT ACCORDANCE WITH THE MANUFACTURER'S INSTALLATION INSTRUCTIONS. SKIMMERS SHALL BE INSTALLED ON THE BASIS OF ONE PER 800 SQUARE FEET OF SURFACE AREA OR FRACTION THEREOF, AND SHALL BE DESIGNED FOR A FLOW RATE OF AT LEAST 25 GALLONS PER MINUTE (GPM) PER SKIMMER. APPROVED MANUFACTURED INLET FITTINGS FOR THE RETURN OF RECIRCULATED POOL WATER SHALL BE PROVIDED ON THE BASIS OF AT LEAST ONE PER 300 SQUARE FEET OF SURFACE AREA. SUCH INLET FITTINGS SHALL BE DESIGNED AND CONSTRUCTED TO ENSURE AN ADEQUATE SEAL TO THE POOL STRUCTURE AND SHALL INCORPORATE A CONVENIENT MEANS OF SEALING FOR PRESSURE TESTING OF THE POOL CIRCULATION PIPING. WHERE MORE THAN ONE INLET IS REQUIRED, THE SHORTEST DISTANCE BETWEEN ANY TWO REQUIRED INLETS SHALL BE AT LEAST 10 FEET.

12. ALL EXPOSED PIPING AND VALVES TO BE CLEARLY LABELED TO INDICATE FUNCTION AND FLOW DIRECTION. PLASTIC PIPE SUBJECT TO A PERIOD OF PROLONGED SUNLIGHT EXPOSURE SHALL BE COATED TO PROTECT IT FROM ULTRAVIOLET LIGHT DEGRADATION. ALL PLASTIC PIPE USED IN THE RECIRCULATION SYSTEM SHALL BE IMPRINTED WITH THE MANUFACTURER'S NAME AND THE NSF-PW LOGO FOR POTABLE WATER APPLICATIONS. RETURN LINES, MAIN DRAIN LINES, AND SURFACE OVERFLOW SYSTEM LINES, SHALL EACH HAVE PROPORTIONING VALVES.

13. ALL POOL PIPING SHALL BE TESTED AND PROVED TIGHT UNDER A STATIC WATER OR AIR PRESSURE TEST OF NOT LESS THAN 35 POUNDS PER SQUARE INCH (PSI) FOR 15 MINUTES. ALL DRAIN AND WASTE PIPING SHALL BE TESTED BY FILLING WITH WATER TO THE POINT OF OVERFLOW AND ALL JOINTS SHALL BE TIGHT. RESULTS OF THIS TESTING SHALL BE DOCUMENTED AND PROVIDED FOR REFERENCE.

14. AN APPROVED HYDROSTATIC RELIEF DEVICE SHALL BE INSTALLED AT THE LOCATION OF HIGH GROUNDWATER TABLE.

15. ELECTRICAL EQUIPMENT WIRING AND INSTALLATION, INCLUDING THE BONDING AND GROUNDING OF POOL COMPONENTS SHALL COMPLY WITH CHAPTER 27 OF THE FLORIDA BUILDING CODE, BUILDING. OUTLETS SUPPLYING POOL PUMP MOTORS CONNECTED TO SINGLE-PHASE 120-VOLT THROUGH 240-VOLT BRANCH CIRCUITS, WHETHER BY RECEPTACLE OR BY DIRECT CONNECTION, AND OUTLETS SUPPLYING OTHER ELECTRICAL EQUIPMENT AND UNDERWATER LUMINAIRES OPERATING AT VOLTAGES GREATER THAN THE LOW VOLTAGE CONTACT LIMIT, CONNECTED TO SINGLE-PHASE, 120 VOLT THROUGH 240 VOLT BRANCH CIRCUITS, RATED 15 OR 20 AMPERES, WHETHER BY RECEPTACLE OR BY DIRECT CONNECTION, SHALL BE PROVIDED WITH GROUND-FAULT CIRCUIT INTERRUPTER PROTECTION FOR PERSONNEL. ALL ELECTRICAL WORK TO BE DESIGNED AND PERFORMED BY OTHERS.

16. ALL ELECTRICALLY ENERGIZED EQUIPMENT SHALL BE EQUIPOTENTIAL GROUND BONDED USING A MINIMUM OF #8 AWG COPPER AND LISTED CLAMPS. IMPENDENCE BETWEEN ANY TWO EQUIPOTENTIAL POINTS SHALL NOT EXCEED 2 OHMS.

17. NO OVERHEAD WIRES SHALL BE LOCATED WITHIN 15'-0" OF POOL.

18. POOL LIGHT NICHES CONTAINING METAL SHALL BE BONDED TO REINFORCING STEEL WITH #8 AWG COPPER WIRE. WIRING FROM LIGHT TO THE JUNCTION BOX SHALL BE RUN IN AN NEC APPROVED PVC CONDUIT UNDER NEC REGULATIONS.

19. SYSTEM TOTAL DYNAMIC HEAD CALCULATIONS ARE PERFORMED BASED ON AN ASSUMED PIPE LAYOUT AS WELL AS A NEGLIGIBLE ELEVATION VARIATION. THE CONTRACTOR SHALL PROVIDE A FIELD DRAWING OF THE FINAL PIPE LAYOUT AND THE PRECISE NUMBER AND TYPE OF FITTINGS USED PRIOR TO INSTALLATION OF THE PUMP SO THE ENGINEER CAN VALIDATE THE CALCULATED TOTAL DYNAMIC HEAD. IF FIELD CONDITIONS RESULT IN AN ELEVATION VARIATION BETWEEN THE POOL AND THE POOL EQUIPMENT, IT SHALL BE REPORTED TO THE ENGINEER IN WRITING PRIOR TO INSTALLING THE PUMP TO ALLOW THE ENGINEER TO VALIDATE THE CALCULATED TOTAL DYNAMIC HEAD.

20. ANY EXISTING IMPERVIOUS AREA OR ISR CALCULATIONS SHALL BE FIELD VERIFIED BY THE CONTRACTOR AND PROVIDED TO THE ENGINEER IN WRITING PRIOR TO CONSTRUCTION.

21. ALL SITE ELEVATIONS ARE OBTAINED FROM THE SITE SURVEY PROVIDED BY THE CONTRACTOR. CONTRACTOR SHALL FIELD VERIFY ALL ELEVATIONS IN THE FIELD AND REPORT ANY DEVIATIONS TO THE ENGINEER IN WRITING PRIOR TO PROCEEDING.

RESIDENTIAL SWIMMING POOL BARRIER REQUIREMENT:

A NEW PROPOSED FENCE OR SCREEN ENCLOSURE WILL BE INSTALLED UNDER A SEPARATE PERMIT THAT CONFORMS WITH THE NECESSARY REGULATIONS NOTED BELOW:

R4501.17.1.1
THE TOP OF THE BARRIER SHALL BE AT LEAST 48 INCHES (1219 MM) ABOVE GRADE MEASURED ON THE SIDE OF THE BARRIER WHICH FACES AWAY FROM THE SWIMMING POOL. THE MAXIMUM VERTICAL CLEARANCE BETWEEN GRADE AND THE BOTTOM OF THE BARRIER SHALL BE 2 INCHES (51 MM) MEASURED ON THE SIDE OF THE BARRIER WHICH FACES AWAY FROM THE SWIMMING POOL. WHERE THE TOP OF THE POOL STRUCTURE IS ABOVE GRADE THE BARRIER MAY BE AT GROUND LEVEL OR MOUNTED ON TOP OF THE POOL STRUCTURE. WHERE THE BARRIER IS MOUNTED ON TOP OF THE POOL STRUCTURE, THE MAXIMUM VERTICAL CLEARANCE BETWEEN THE TOP OF THE POOL STRUCTURE AND THE BOTTOM OF THE BARRIER SHALL BE 4 INCHES (102 MM).

R4501.17.1.2
THE BARRIER MAY NOT HAVE ANY GAPS, OPENINGS, INDENTATIONS, PROTRUSIONS, OR STRUCTURAL COMPONENTS THAT COULD ALLOW A YOUNG CHILD TO CRAWL UNDER, SQUEEZE THROUGH, OR CLIMB OVER THE BARRIER AS HEREIN DESCRIBED BELOW. ONE END OF A REMOVABLE CHILD BARRIER SHALL NOT BE REMOVABLE WITHOUT THE AID OF TOOLS. OPENINGS IN ANY BARRIER SHALL NOT ALLOW PASSAGE OF A 4-INCH-DIAMETER (102 MM) SPHERE.

R4501.17.1.3
SOLID BARRIERS WHICH DO NOT HAVE OPENINGS SHALL NOT CONTAIN INDENTATIONS OR PROTRUSIONS EXCEPT FOR NORMAL CONSTRUCTION TOLERANCES AND TOOLED MASONRY JOINTS.

R4501.17.1.4
WHERE THE BARRIER IS COMPOSED OF HORIZONTAL AND VERTICAL MEMBERS AND THE DISTANCE BETWEEN THE TOPS OF THE HORIZONTAL MEMBERS IS LESS THAN 45 INCHES (1143 MM), THE HORIZONTAL MEMBERS SHALL BE LOCATED ON THE SWIMMING POOL SIDE OF THE FENCE. SPACING BETWEEN VERTICAL MEMBERS SHALL NOT EXCEED 13/4 INCHES (44 MM) IN WIDTH. WHERE THERE ARE DECORATIVE CUTOUTS WITHIN VERTICAL MEMBERS, SPACING WITHIN THE CUTOUTS SHALL NOT EXCEED 13/4 INCHES (44 MM) IN WIDTH.

R4501.17.1.5
WHERE THE BARRIER IS COMPOSED OF HORIZONTAL AND VERTICAL MEMBERS AND THE DISTANCE BETWEEN THE TOPS OF THE HORIZONTAL MEMBERS IS 45 INCHES (1143 MM) OR MORE, SPACING BETWEEN VERTICAL MEMBERS SHALL NOT EXCEED 4 INCHES (102 MM). WHERE THERE ARE DECORATIVE CUTOUTS WITHIN VERTICAL MEMBERS, SPACING WITHIN THE CUTOUTS SHALL NOT EXCEED 13/4 INCHES (44 MM) IN WIDTH.

R4501.17.1.6
MAXIMUM MESH SIZE FOR CHAIN LINK FENCES SHALL BE A 21/4-INCH SQUARE (57 MM) UNLESS THE FENCE IS PROVIDED WITH SLATS FASTENED AT THE TOP OR BOTTOM WHICH REDUCE THE OPENINGS TO NO MORE THAN 13/4 INCHES (44 MM).

R4501.17.1.7
WHERE THE BARRIER IS COMPOSED OF DIAGONAL MEMBERS, THE MAXIMUM OPENING FORMED BY THE DIAGONAL MEMBERS SHALL BE NO MORE THAN 13/4 INCHES (44 MM).

R4501.17.1.8
ACCESS GATES, WHEN PROVIDED, SHALL BE SELF-CLOSING AND SHALL COMPLY WITH THE REQUIREMENTS OF SECTIONS R4501.17.1.1 THROUGH R4501.17.1.7 AND SHALL BE EQUIPPED WITH A SELF-LATCHING LOCKING DEVICE LOCATED ON THE POOL SIDE OF THE GATE. WHERE THE DEVICE RELEASE IS LOCATED NO LESS THAN 54 INCHES (1372 MM) FROM THE BOTTOM OF THE GATE, THE DEVICE RELEASE MECHANISM MAY BE LOCATED ON EITHER SIDE OF THE GATE AND SO PLACED THAT IT CANNOT BE REACHED BY A YOUNG CHILD OVER THE TOP OR THROUGH ANY OPENING OR GAP FROM THE OUTSIDE. GATES THAT PROVIDE ACCESS TO THE SWIMMING POOL MUST OPEN OUTWARD AWAY FROM THE POOL. THE GATES AND BARRIER SHALL HAVE NO OPENING GREATER THAN 1/2 INCH (12.7 MM) WITHIN 18 INCHES (457 MM) OF THE RELEASE MECHANISM.

Vito A Gironda — Digitally signed by Vito A Gironda Date: 2025.09.04 16:06:26 -04'00'

DATE: 1/15/2025
PROJECT NO.
LOT 10 OXFORD PRESERVE TRL OXFORD, FL

REVISION	DATE
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VITO GIRONDA ENGINEERING, LLC
DESIGNS@VITOGIRONDAENGINEERING.COM | (330) 807-8643
8499 SERRANO CIRCLE
MELBOURNE, FL 32940

RESIDENTIAL SWIMMING POOL DESIGN
Lot 10 Oxford Preserve Trl Oxford, FL
ISSUED FOR APPROVAL

I HEREBY CERTIFY THAT THIS ENGINEERING DOCUMENT WAS PREPARED BY ME OR UNDER MY DIRECT SUPERVISION AND THAT I AM A DULY LICENSED PROFESSIONAL ENGINEER UNDER THE LAWS OF THE STATE OF FLORIDA.

THIS ITEM HAS BEEN DIGITALLY SIGNED AND SEALED BY VITO GIRONDA, PE ON THE DATE ADJACENT TO THE SEAL. PRINTED COPIES OF THIS DOCUMENT ARE NOT CONSIDERED SIGNED AND SEALED AND THE SIGNATURE MUST BE VERIFIED ON ANY ELECTRONIC COPIES.

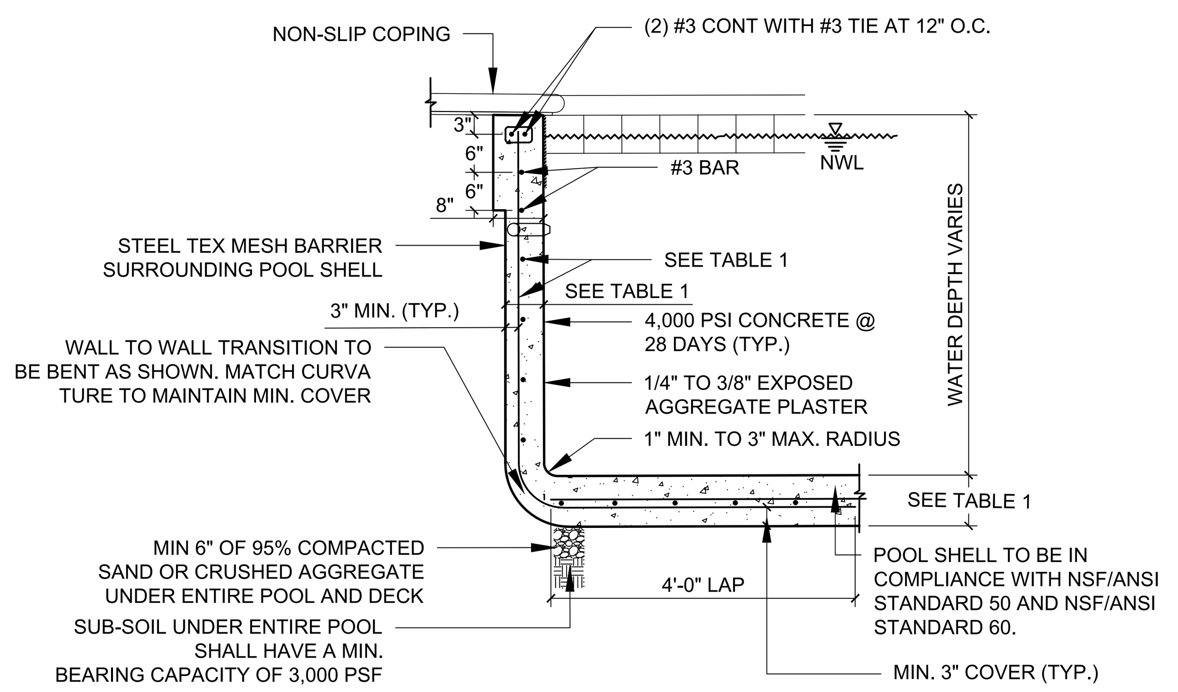
VITO A. GIRONDA LICENSE No 88995 STATE OF FLORIDA PROFESSIONAL ENGINEER

Vito Gironda

VITO GIRONDA
FLORIDA
PROFESSIONAL ENGINEER
NO. 88995

GN1

DRAWN BY: VG
PROJECT: Lot 10 Oxford Preserve Trl Oxford, FL.dwg

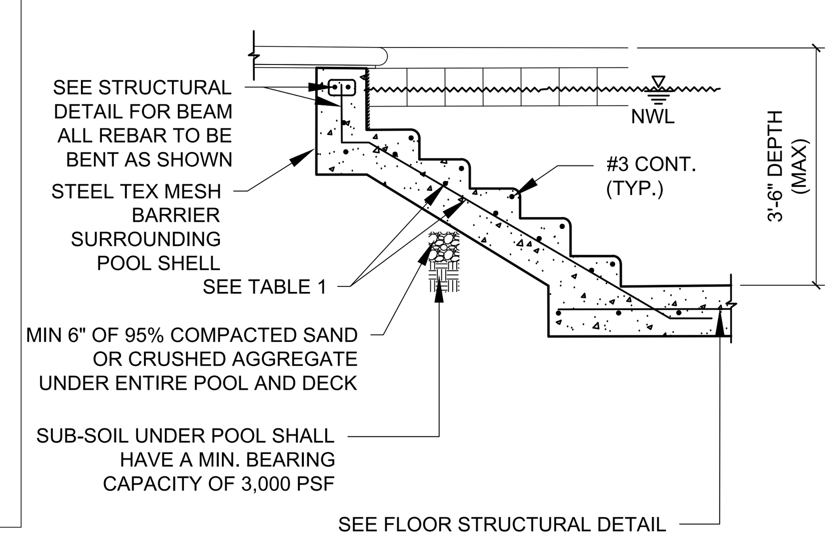


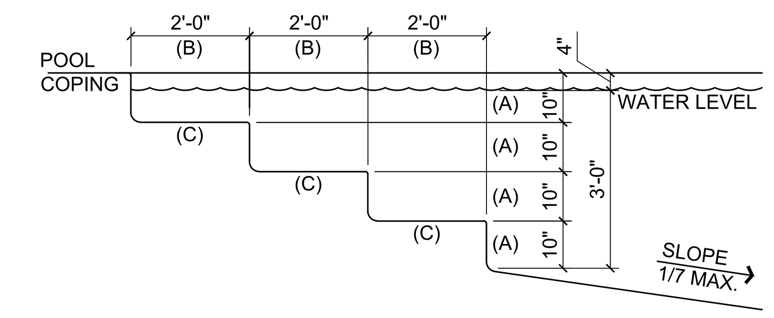
NOTE: OVERLAP ALL REBAR SPLICES 24" UNLESS NOTED OTHERWISE

01 TYP. POOL AND FLOOR SECTION
SCALE: 1" = 2'-0"

NOTE:

1. STEP RISERS SHALL NOT EXCEED 10 INCHES.
2. STEP TREAD DEPTH SHALL BE AT LEAST 12 INCHES.
3. HANDRAILS AND STAIRS SHALL MEET PUBLIC LAW 101-336, 28 CFR PARTS 35 AND 36
4. IF TOP TREAD IS ABOVE WATER LEVEL FINISH WITH SLIP RESISTANT TILE.
5. HANDRAILS TO MATCH SLOPE OF TREADS AT 34" ABOVE THE LEADING EDGE OF TREAD.
6. STRENGTH OF HANDRAILS AND MOUNTING ANCHORS TO MEET 28 CFR PART 36 ADA STANDARDS FOR ACCESSIBLE DESIGN 4.26.3
7. FOR HANDRAIL, ANCHOR AND ESCUTCHEON PLATE INFORMATION SEE MANUFACTURER INFORMATION AND RECOMMENDATIONS.

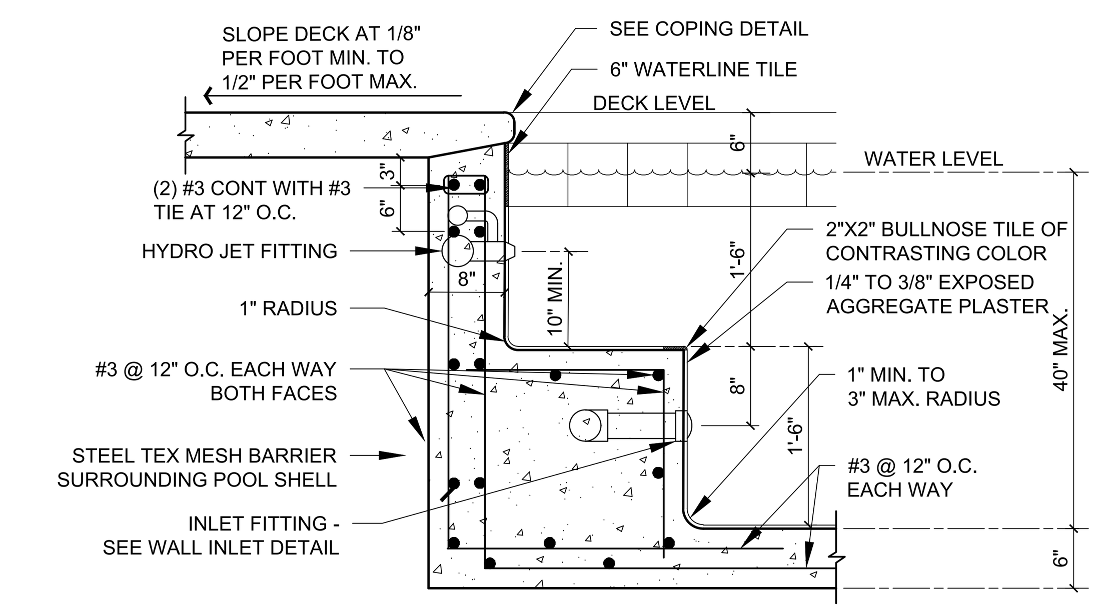




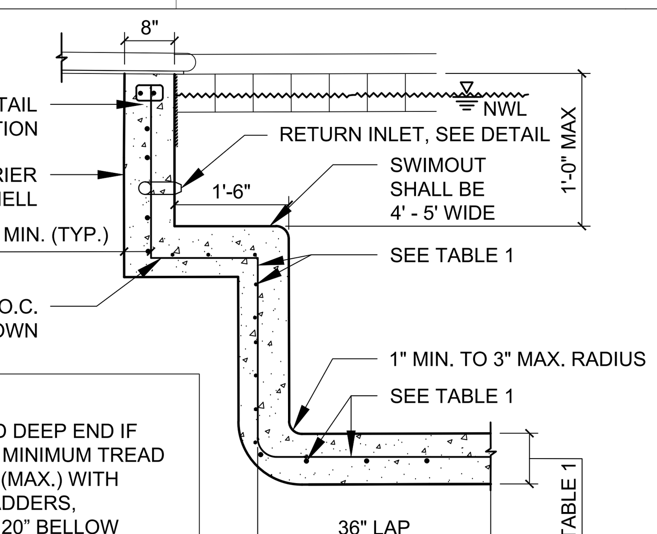
STEP NOTES:
1. RISER HEIGHT MUST BE UNIFORM (12" MAX)
2. CONTRASTING TILE AT EDGE
3. TREADS SHALL HAVE SLIP-RESISTANT SURFACES-ANSI/ASAP-ICC-5 6.2.1
4. ALL STEPS (INCLUDING INTO SPA AND SUNSHELF) SHALL PROVIDE A MIN. OF 240 SQ.IN. SURFACE AREA (TYP.)

02 TYP. POOL STAIR DETAIL
SCALE: 1" = 2'-0"

NOTES:
1. REFER TO TYPICAL WALL SECTION FOR FURTHER INFORMATION.
2. OVERLAP ALL REBAR SPLICES 24" UNLESS NOTED OTHERWISE



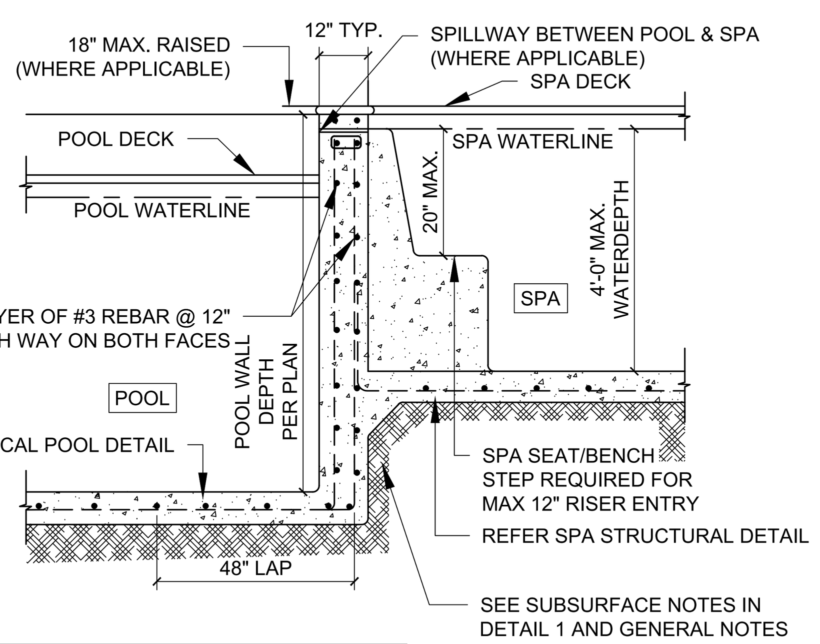
03 SPA BENCH DETAIL
SCALE: 3/4" = 1'-0"



NOTE:
ENTRY/ EXIT REQUIRED AT THE SHALLOW END AND DEEP END IF OVER 5 FEET DEEP. ACCEPTABLE ARE STAIRS (10" MINIMUM TREAD WITH 240 SQUARE INCH MINIMUM AREA, 12" RISER (MAX.) WITH INTERMEDIATE TREADS AND RISERS UNIFORM). LADDERS, UNDERWATER SEATS, AND SWIM-OUTS (MAXIMUM 20" BELLOW WATERLINE). MIN. UNOBSTRUCTED WALKING SURFACE OF 240 SQ. IN. REQUIRED.

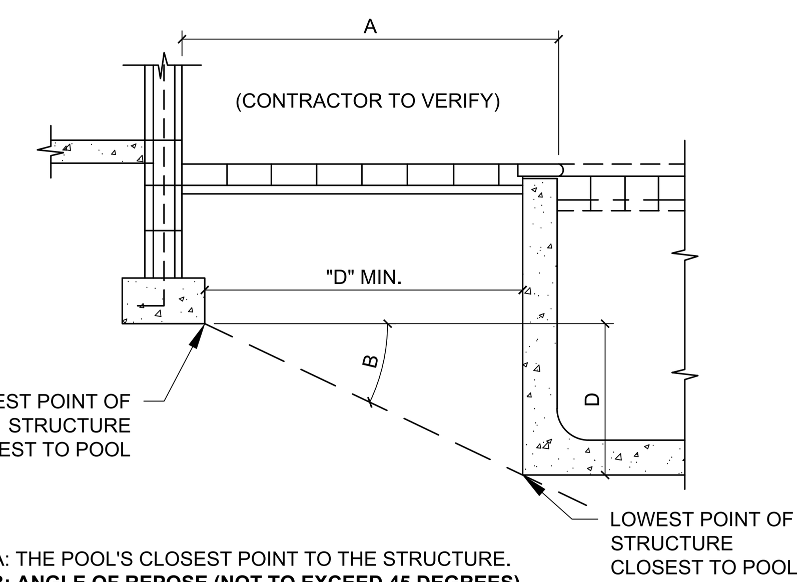
04 TYP. SWIMOUT DETAIL
SCALE: 1" = 2'-0"

NOTE:
SEE DETAIL 3 FOR SPA BEAM CONSTRUCTION



05 TYP. POOL/SPA COMMON WALL
SCALE: 1" = 2'-0"

TABLE 1: TYPICAL POOL WALL/FLOOR THICKNESS & REINFORCEMENT SCHEDULE U.N.O.			
MAX. WATER DEPTH	WALL/FLOOR THICKNESS	WALL/FLOOR REINFORCEMENT SPACING (O.C. EACH WAY) #3 REBAR SPACING (24" MIN. SPLICE)	WALL/FLOOR REINFORCEMENT SPACING (O.C. EACH WAY) #4 REBAR SPACING (24" MIN. SPLICE)
5'-6"	6"	12"	18"
6'-0"	8"	8"	14"
7'-0"	8"	6"	9"



A: THE POOL'S CLOSEST POINT TO THE STRUCTURE.
B: ANGLE OF REPOSE (NOT TO EXCEED 45 DEGREES)

NOTE:
DESIGN IS NOT VALID WHEN ANGLE "B" IS GREATER THAN 45" AS SPECIAL ENGINEERING IS REQUIRED. SEE EXCAVATION NOTE BELOW

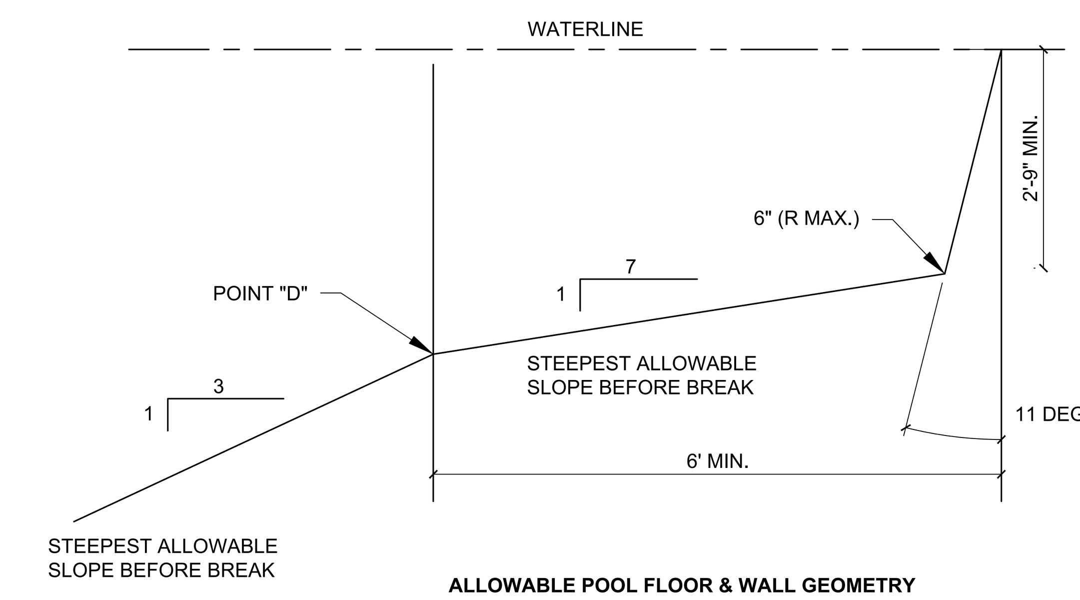
ANGLE OF REPOSE

FBC 1804.1 EXCAVATION NEAR FOUNDATIONS (CONTRACTOR TO PROVIDE):
EXCAVATION FOR ANY PURPOSE SHALL NOT REDUCE VERTICAL OR LATERAL SUPPORT FOR ANY FOUNDATION OR ADJACENT FOUNDATION WIHTOUT FIRST UNDERPINNING OR PROTECTING THE FOUNDATION AGAINST DETERMENTAL LATERAL OR VERTICAL MOVEMENT, OR BOTH.

- THE SLOPE OF THE FLOOR FROM THE SHALLOW END WALL TOWARDS THE DEEP AREA SHALL NOT EXCEED A 1:7 INCLINE TO THE POINT OF THE FIRST SLOPE CHANGE.
- CHANGES IN THE SLOPE BETWEEN SHALLOW AND DEEP AREAS SHALL BE AT A MINIMUM WATER DEPTH OF 2 FT. 9 IN. AND BE AT LEAST 6 FT. FROM SHALLOW END, EXCEPT AS SPECIFIED IN PARA. 6.3. ANSI/APSP/ICC-5 "SHALLOW END DETAIL FOR BEACH AND SLOPING ENTRIES".
- THE SLOPE OF THE FLOOR SHALL NOT EXCEED A 1:3 INCLINE FOR A DEEPER AREA AFTER 1:7 INCLINATION

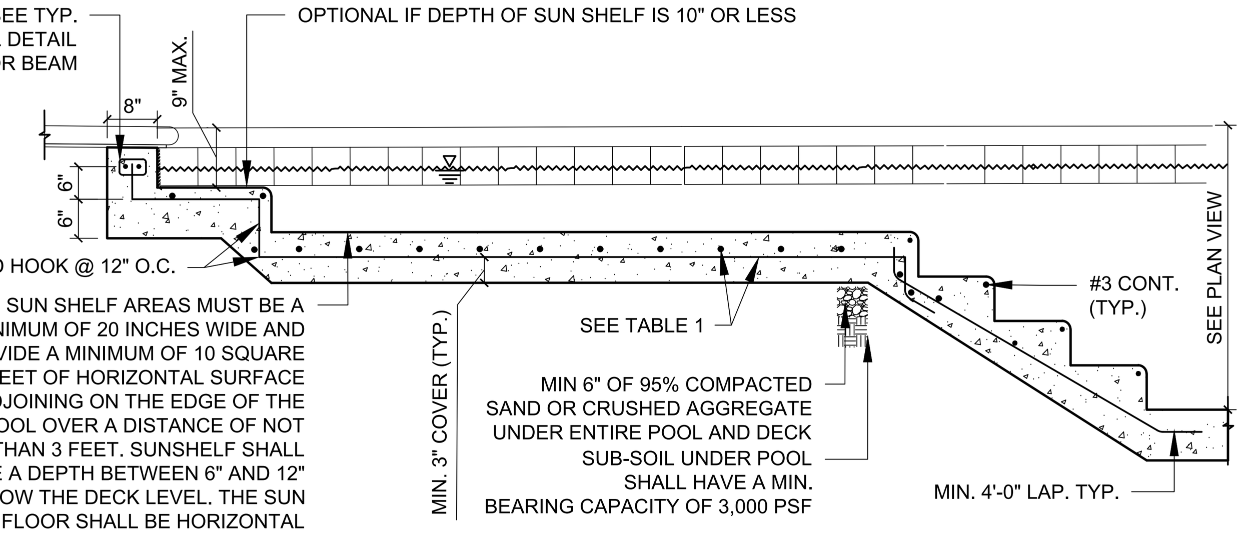
ANSI/ APSP - 5 SECTION 17.2 ROPE AND FLOAT:

- IN POOLS WHERE THE POINT OF THE FIRST SLOPE CHANGE (POINT D, FIGURE BELOW) OCCURS IN A WATER DEPTHS LESS THAN 4'-6", A ROPE AND A FLOAT ASSEMBLY SHALL BE INSTALLED ACROSS THE WIDTH OF THE POOL GENERALLY PARALLEL TO, AND AT A MIN. OF 1 FT. AND A MAX. OF 2 FT. ON THE SHALLOW SIDE OF THE CHANGE IN FLOOR SLOPE.



ALLOWABLE POOL FLOOR & WALL GEOMETRY

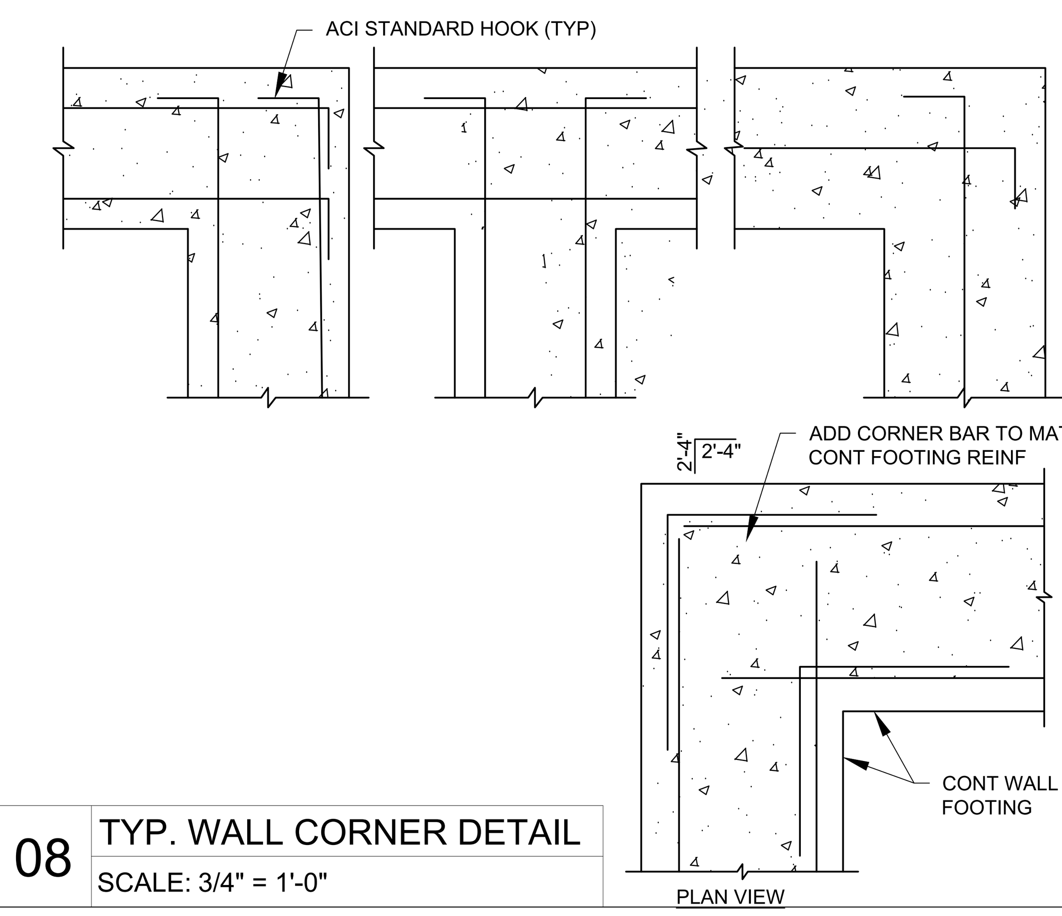
06 POOL GEOMETERY
SCALE: NTS



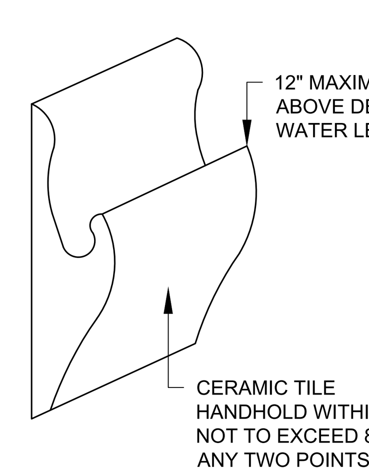
SUN SHELF AREAS MUST BE A MINIMUM OF 20 INCHES WIDE AND PROVIDE A MINIMUM OF 10 SQUARE FEET OF HORIZONTAL SURFACE ADJOINING ON THE EDGE OF THE POOL OVER A DISTANCE OF NOT LESS THAN 3 FEET. SUNSHELF SHALL HAVE A DEPTH BETWEEN 6" AND 12" BELOW THE DECK LEVEL. THE SUN SHELF FLOOR SHALL BE HORIZONTAL OR SLOPED AT A MAXIMUM OF 1 UNIT VERTICAL IN 60 UNITS HORIZONTAL.

NOTE: OVERLAP ALL REBAR SPLICES 24" UNLESS NOTED OTHERWISE

07 TYP. SUNSHELF DETAIL
SCALE: 1" = 2'-0"



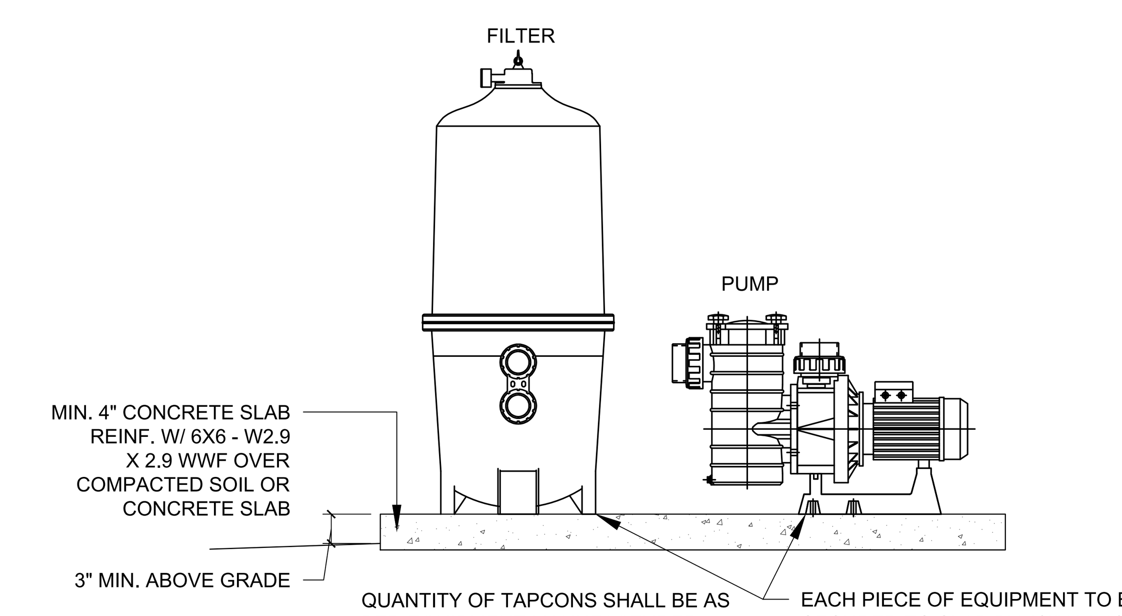
08 TYP. WALL CORNER DETAIL
SCALE: 3/4" = 1'-0"



NOTE: HANDHOLD SHALL BE PROVIDED AROUND THE POOL EDGE IN ANY AREA WHERE THE WATER DEPTH EXCEEDS 4 FT. AS PER ANSI/ APSP -5 SECTION 17.1

NSPI-5-2003
15.1.1 - HANDHOLDS SHALL BE ACCESSIBLE WITHIN 4' NOT TO EXCEED 8' AT ANY TWO POINTS ALONG THE INSIDE OF THE POOL PERIMETER WHERE THE WATER DEPTH EXCEEDS 4'

09 TYP. HANDHOLD DETAIL
SCALE: NTS



- CONTRACTOR TO FOLLOW EQUIPMENT MANUFACTURER'S ANCHORING INSTRUCTIONS. ALL EQUIPMENT SHALL BE ANCHORED TO RESIST WIND LOADS OF 150 MPH (DESIGNED BY OTHERS)
- ANY POOL EQUIPMENT SHOULD BE SECURELY ANCHORED TO A PLATFORM, THE ELEVATION OF THE PLATFORM SHOULD BE TO OR ABOVE THE BASE FLOOD ELEVATION ESTABLISHED FOR THE ZONE.
- PRE-CAST HURRICANE PADS ARE AN ACCEPTABLE ALTERNATIVE

10 EQUIPMENT ANCHORING
SCALE: NTS

DATE: 1/15/2025
PROJECT NO.
LOT 10 OXFORD PRESERVE TRL OXFORD, FL

REVISION	DATE
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VITO GIRONDA ENGINEERING, LLC
DESIGNS@VITOGIRONDAENGINEERING.COM | (330) 807-8643
8499 SERRANO CIRCLE
MELBOURNE, FL 32940

VGE

RESIDENTIAL SWIMMING POOL DESIGN
Lot 10 Oxford Preserve Trl Oxford, FL
ISSUED FOR APPROVAL

I HEREBY CERTIFY THAT THIS ENGINEERING DOCUMENT WAS PREPARED BY ME OR UNDER MY DIRECT SUPERVISION AND THAT I AM A DULY LICENSED PROFESSIONAL ENGINEER UNDER THE LAWS IN THE STATE OF FLORIDA.

THIS ITEM HAS BEEN DIGITALLY SIGNED AND SEALED BY VITO GIRONDA, PE ON THE DATE ADJACENT TO THE SEAL. SIGNATURE MUST BE VERIFIED ON ANY ELECTRONIC COPIES. ANY PRINTED COPIES ARE NOT CONSIDERED SIGNED AND SEALED.

VITO A. GIRONDA LICENSE No 88995 STATE OF FLORIDA PROFESSIONAL ENGINEER

Vito Gironda

VITO GIRONDA
FLORIDA
PROFESSIONAL ENGINEER
NO. 88995

D1

DRAWN BY: VG
PROJECT: Lot 10 Oxford Preserve Trl Oxford, FL.dwg

ELECTRICAL REQUIREMENTS:

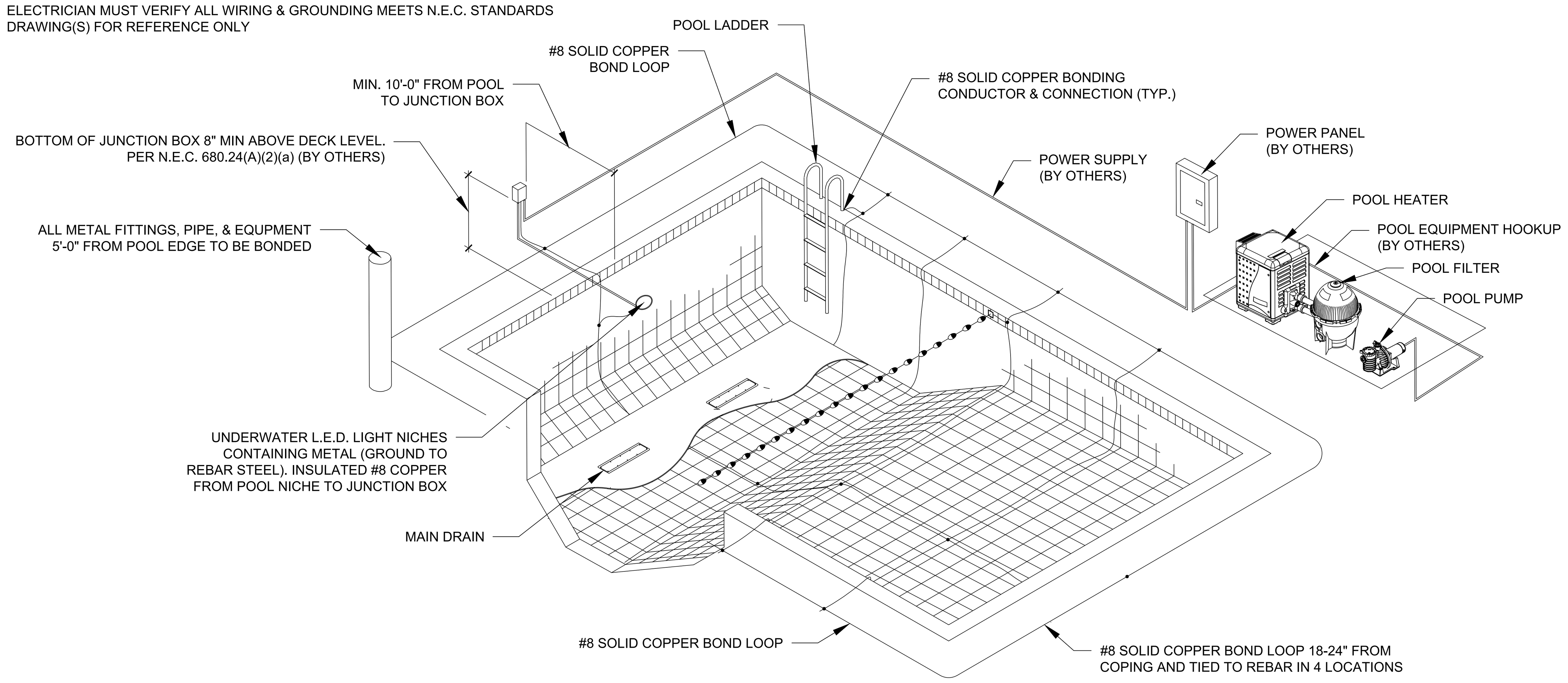
WIRING AND BONDING AND ALL ELECTRICAL SHALL CONFORM TO NEC ARTICLE 680.
NO OUTLET OR OVERHEAD POWER WITHIN 10', IF WITHIN 15' PROTECT WITH GFI.
BRASS FITTINGS TO J-BOX OR TRANSFORMER WHICHEVER IS FIRST, EXCEPT WHERE PVC IS APPROVED.
BONDING GRID PER NEC 680.26 (B) (2) (B) ALTERNATE MEANS EQUIPOTENTAL BONDING CONDUCTOR MUST FOLLOWING REQUIREMENTS:

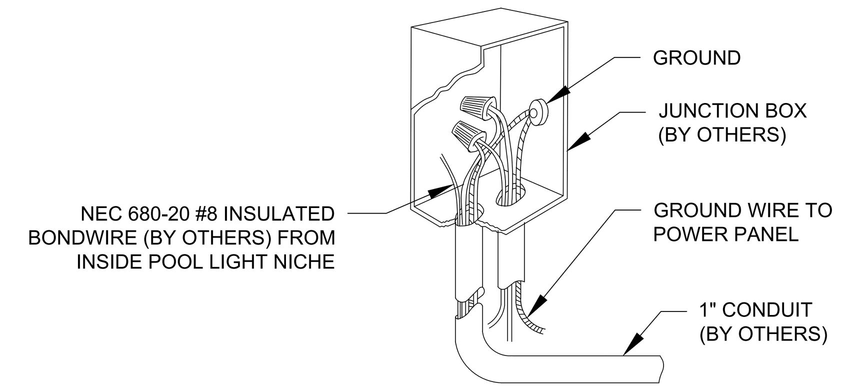
(1) 8 AWS BARE SOLID COPPER BONDING CONDUCTOR
(2) THE BONDING CONDUCTOR MUST FOLLOW THE CONTOUR OF THE PERIMETER SURFACE
(3) LISTED SPLICING DEVICES
(4) BONDING CONDUCTOR MUST BE 18 TO 24 INC. FROM INSIDE WALLS OF THE POOL
(5) BONDING CONDUCTOR MUST BE WITHIN OR UNDER THE PERIMETER SURFACE 3 TO 6 IN. BELOW THE SUBGRADE.

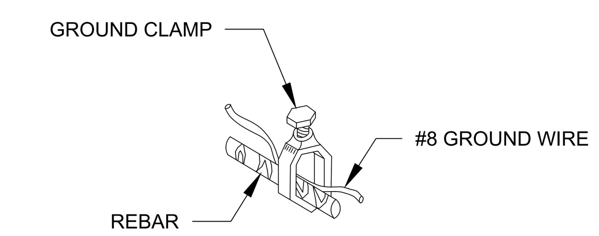
ALL CONTROL ENCLOSURES (INCLUDING ACCESSORY ELECTRONIC EQUIPMENT WITH METAL ENCLOSURES), PUMP MOTORS, HEATERS, LIGHTS, LIGHT TRANSFORMERS, HANDRAIL AND LADDER DECK ANCHORS, ANY METAL, WINDOW AND DOOR FRAMES WITHIN 5 FEET OR LESS OF THE WATER'S EDGE ARE TO BE GROUNDED IN ACCORDANCE WITH ARTICLE 680 OD THE 2020 NEC.

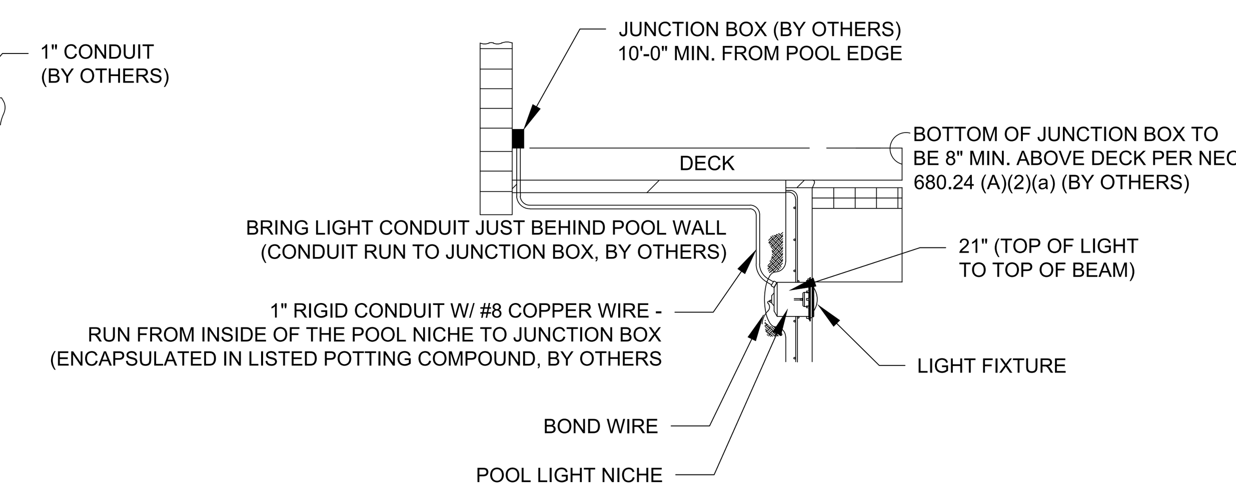
ALL POOL PUMP MOTOR(S) TO HAVE GFCI PROTECTION (NEC 680.22 (B), FBC SECTION 27), AND FRC 4101.16

- ALL GROUNDING BY ELECTRICIAN
- ALL ELECTRICAL FIXTURES TO BE U.L. APPROVED
- ELECTRICIAN MUST VERIFY ALL WIRING & GROUNDING MEETS N.E.C. STANDARDS
- DRAWING(S) FOR REFERENCE ONLY

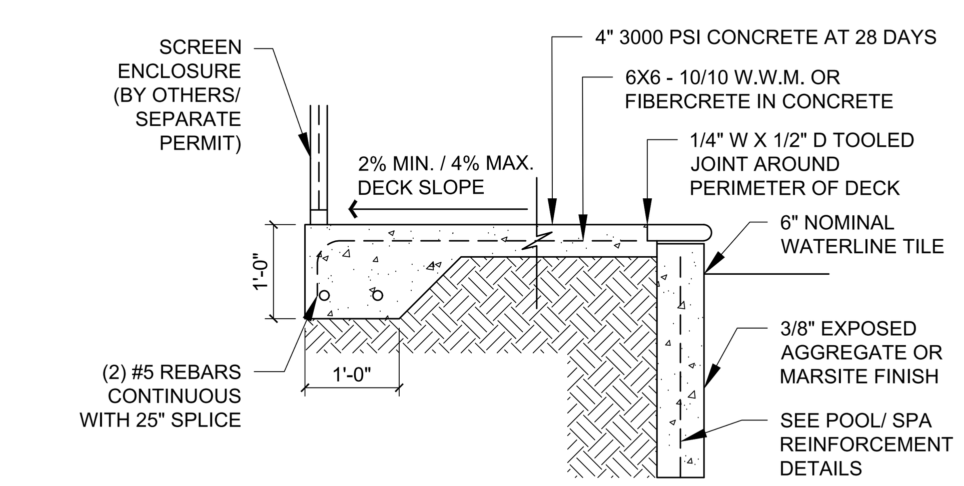




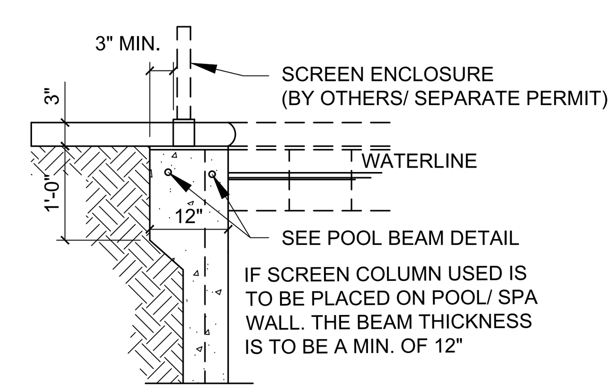




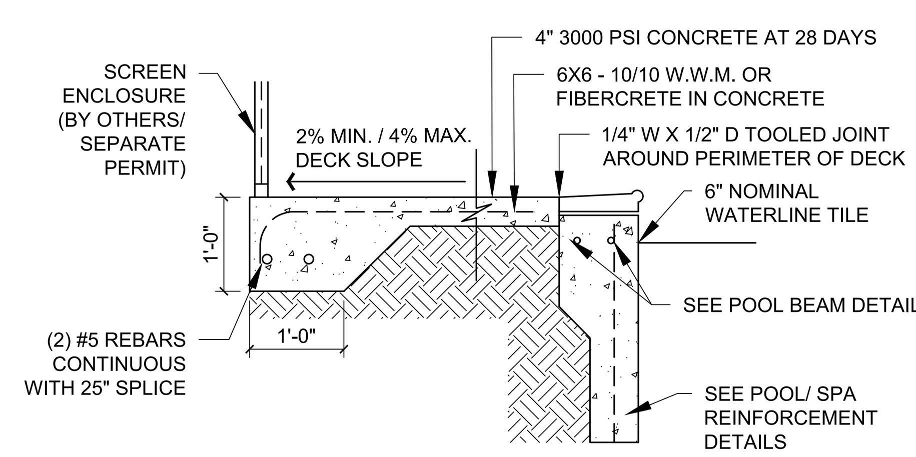
01 BONDING PLAN
SCALE: 1" = 7'-0"



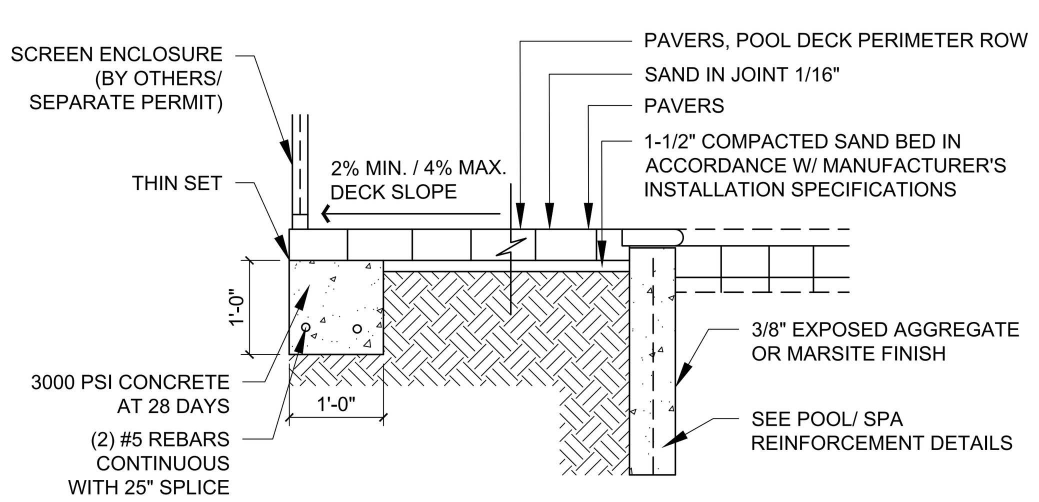
02 POOL/ SPA CONC. DECK DETAIL
SCALE: NTS



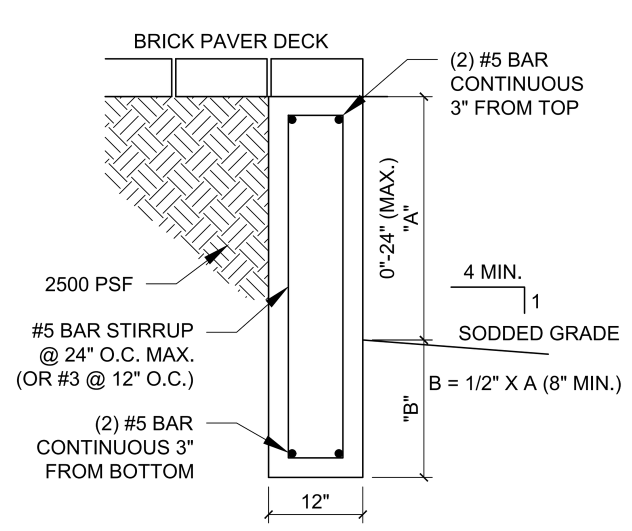
03 SCREEN ENCLOSURE ON POOL BEAM DETAIL
SCALE: NTS



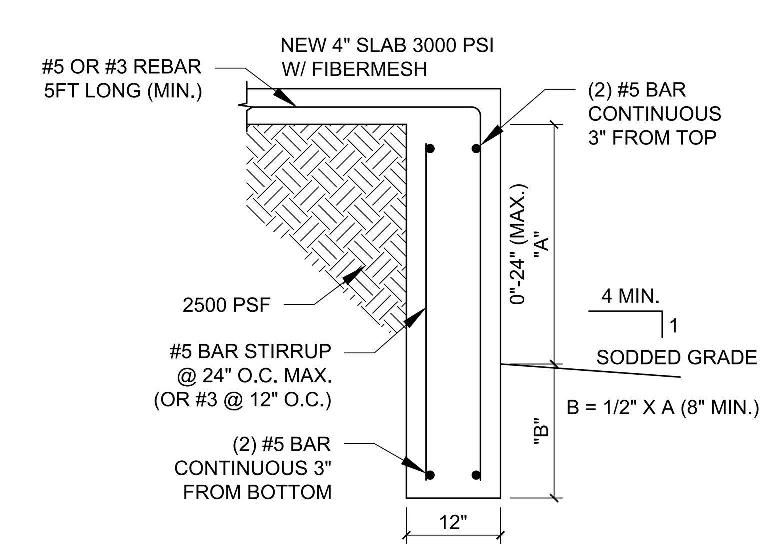
04 COPING POOL/SPA. DECK AND BEAM DETAIL
SCALE: NTS



05 INTERLOCKING PAVERS, FOOTING AND TYP. DECK DETAIL
SCALE: NTS



06 TURNDOWN FOOTER DETAIL (0 TO 24 INCHES ABOVE GRADE)
SCALE: NTS



07 TURNDOWN FOOTER DETAIL (0 TO 24 INCHES ABOVE GRADE)
SCALE: NTS

DESIGN CRITERIA:
APPLICABLE CODES, REGULATIONS & STANDARDS
1. THE 2023 FLORIDA BUILDING CODE, SPECIFICALLY CHAPTER 16 STRUCTURE DESIGN & CH. 19 CONCRETE.
2. AA ALUMINUM DESIGN MANUAL 2020.
3. ASCE/ SEI 7-22
4. NDS NATIONAL DESIGN SPECIFICATION FOR WOOD.
5. ACI 318 CONCRETE REFERENCE MANUAL.

SPECIFICATIONS:
THE FOLLOWING SPECIFICATIONS ARE APPLICABLE TO THIS PROJECT:
1. WHERE CONCRETE SPECIFICATIONS ARE REQUIRED, WHETHER IN THE SCREEN ENCLOSURE SCOPE OR NOT, BY ONE OR MORE REGULATORY AGENCY, THE FOLLOWING SPECIFICATIONS ARE APPLICABLE:
A. CONCRETE SHALL CONFORM TO ASTM C94 FOR THE FOLLOWING COMPONENTS:
I. PORTLAND CEMENT TYPE 1 - ASTM C 150
II. AGGREGATES - LARGE AGGREGATE 3/4 MAX. - ASTM C 33
III. AIR ENTRAINING +/- 1% - ASTM C 260
Iv. WATER REDUCING AGENT - ASTM C 494
v. CLEAN POTABLE WATER
VI. OTHER ADMIXTURES NOT PERMITTED
B. METAL ACCESSORIES SHALL CONFORM TO:
I. REINFORCING BARS - ASTM A615, GRADE 60
II. WELDED WIRE FABRIC - ASTM A185
C. CONCRETE SLUMP AT DISCHARGE CHUTE NOT LESS THAN 3" OR MORE THAN 5". WATER ADDED AFTER BATCHING IS NOT PERMITTED.
D. PREPARE & PLACE CONCRETE PER AMERICAN CONCRETE INSTITUTE MANUEL OF STANDARD PRACTICE, PART 1,2 & 3 INCLUDING NOT WEATHER RECOMMENDATIONS.
E. MOIST CURE OR POLYETHYLENE CURING PERMITTED.
F. PRIOR TO PLACING CONCRETE, TREAT THE ENTIRE SUBSURFACE AREA FOR TERMITES IN COMPLIANCE WITH THE FBC. FOR RISK CATEGORY II, III & IV, STRUCTURES ONLY.
G. CONCRETE SLAB SHALL BE PLACES OVER A POLYETHYLENE VAPOR BARRIER. (SLAB ONLY)
2. SITE SPECIFIC ENGINEERING REQUIRED TO DETERMINE FOUNDATION SUFFICIENCY FOR STRUCTURES ANCHORED TO FOUNDATION.
3. WHEN PAVERS ARE UNDER ALUMINUM MEMBERS, CONTRACTOR SHALL EPOXY TO DECK OR GROUT TO DECK w/3000 PSI GROUT WITH BONDING AGENT
4. WHEN APPLICABLE FOR NEW SLAB ADDITION TO ADJACENT DRILL & EPOXY #4 X 8" REBAR INTO EX. FOUNDATION EMBED 4" MIN W/ NON-SHRINKING SIMPSON EPOXY-TIE (OR EQUAL) 48" O.C. TYP. ALL LOCATIONS
5. WHEN APPLICABLE FOR NEW FOOTER TO EXISTING, DRILL & EPOXY NEW STEEL INTO EX. FOUNDATION WITH EMBED 6" MIN W/ NON-SHRINKING SIMPSON EPOXY-TIE (OR EQUAL) TYP. ALL LOCATIONS
6. WHERE PAVERS ARE UNDER ALUMINUM MEMBERS, CONTRACTOR SHALL EPOXY TO DECK OR GROUT TO DECK w/3000 PSI GROUT WITH BONDING AGENT.
7. MINIMUM CONCRETE STRENGTH 3000 PSI UNLESS OTHERWISE NOTED.

MASONRY SPECIFICATIONS
1. CONCRETE MASONRY UNITS (CMU) SHALL BE STANDARD HOLLOW UNITS AND SHALL BE 1900 PSI MINIMUM BASED ON TYPE M OR S MORTAR.
2. ALL MORTAR SHALL BE TYPE M OR S.
3. ALL GROUT SHALL BE 2000 PSI MINIMUM AND HAVE MAXIMUM COARSE AGGREGATE SIZE OF 3/8".
4. PROVIDE CLEAN-OUTS FOR REINFORCED CELLS CONTAINING REINFORCEMENT WHEN GROUT POUR EXCEEDS 5'-0" IN HEIGHT.

FOOTING NOTES
1. PROVIDE 1-1/2" COVERAGE TOP, SIDES, BOTTOM AND 1" BETWEEN ADJACENT REBAR LAPS.
2. PROVIDE MIN. 3" COVERAGE OF REBAR FOR ALL CONCRETE IN CONTACT WITH THE EARTH.
3. FOOTING CONCRETE SHALL BE MIN. 3000 PSI AT 28 DAYS
4. FOOTING REINFORCEMENT SHALL BE MIN. GRADE 60
5. PROVIDE POSITIVE DRAINAGE AWAY FROM STRUCTURE TO CITY / CO. REQUIREMENTS
6. PROVIDE 2500 PSF BEARING (TYPICAL) UNDER FOUNDATION
7. SEE GENERAL NOTES ABOVE FOR TYING INTO EXISTING FOUNDATIONS
8. SEE GENERAL NOTES ABOVE FOR ADDITIONAL CONCRETE INFORMATION & SPECS.

DATE: 1/15/2025
PROJECT NO.
LOT 10 OXFORD PRESERVE TRL OXFORD, FL

REVISION	DATE
1	
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VITO GIRONDA ENGINEERING, LLC
DESIGNS@VITOGIRONDAENGINEERING.COM | (330) 807-8643
8499 SERRANO CIRCLE
MELBOURNE, FL 32940

VGE

RESIDENTIAL SWIMMING POOL DESIGN
Lot 10 Oxford Preserve Trl Oxford, FL
ISSUED FOR APPROVAL

I HEREBY CERTIFY THAT THIS ENGINEERING DOCUMENT WAS PREPARED BY ME OR UNDER MY DIRECT SUPERVISION AND THAT I AM A DULY LICENSED PROFESSIONAL ENGINEER UNDER THE LAWS IN THE STATE OF FLORIDA.

THIS ITEM HAS BEEN DIGITALLY SIGNED AND SEALED BY VITO GIRONDA, PE ON THE DATE ADJACENT TO THE SEAL. SIGNATURE MUST BE VERIFIED ON ANY ELECTRONIC COPIES. ANY PRINTED COPIES ARE NOT CONSIDERED SIGNED AND SEALED.

VITO A. GIRONDA
LICENSE
No 88995
STATE OF FLORIDA
PROFESSIONAL ENGINEER

Vito Gironda

VITO GIRONDA
FLORIDA
PROFESSIONAL ENGINEER
NO. 88995

D2

DRAWN BY: VG
PROJECT: Lot 10 Oxford Preserve Trl Oxford, FL.dwg